Many HFs suffer from operational problems, including falling afoul of legal and regulatory requirements. Brown et al. (2012) show that operational risk increases the likelihood of poor performance. Unfortunately, operational risk does not seem to influence the choices asset owners make in selecting HFs, as many investors simply chase the HFs with the best past returns.

4.2. FEES

HF fees are high.

Top-Earning Hedge Fund Managers

Table 17.9 lists the top-earning HF managers in 2011 compiled by *Alpha* magazine. The highest earning HF manager in 2011 was Ray Dalio, who heads Bridgewater Associates. He earned \$3.9 billion. Dalio appeared on the top-earning HF manager list in 2010, where he earned \$3.1 billion. Bridgewater also generated enormous wealth for the two men tied for number 7 on the list, Greg Jensen and Robert Prince. The number 3 manager, James Simons, earned \$2.1 billion in 2011. Simons also appeared on the 2010 and 2009 lists, earning and \$2.5 billion in both years.

You thought the top CEOs were well paid?[40] The right-hand side of Table 17.9 lists compensation of the top-earning CEOs in 2011 compiled by *Forbes*. There is an order of magnitude difference in the compensation of the top CEOs and the top HF managers; we need to multiply the compensation numbers of the top CEOs by at least 10 to make them comparable. Yes, the top-earning HF managers do extremely well, but before immediately rushing out to become a HF manager, remember that these managers are exceptional—the typical HF lifecycle is short, most HFs die after a few years, and you never hear of the worst-paid HF managers.

There are many ways to make the enormous amounts in Table 17.9. Bridgewater, the home of Dalio, Jensen, and Prince, is a macro fund with a quant bent. Renaissance Technologies is a quant firm, with very secretive and technical algorithms. Carl Icahn's fund fits in the event-driven style category and employs many *shareholder activist* techniques, aggressive enough that the market talks of an "Icahn Lift." Brevan Howard is a macro fund, but it runs mostly discretionary (as opposed to quant systematic) portfolios.

Oh, there are no women in Table 17.9.

Hedge Fund Compensation is a Call Option

The top-earning HF managers earn so much because of large management *and* performance fees. The performance fee is an option—the HF manager gets paid

[40] CEO compensation has risen significantly over the past few decades. Gabaix and Landier (2008) argue this is optimal and goes hand in hand with the large increases in market capitalization of the largest companies over this period.